Acute dark chocolate and cocoa ingestion and endothelial function: a randomized controlled crossover trial.

BACKGROUND: Studies suggest cardioprotective benefits of dark chocolate containing cocoa. OBJECTIVE: This study examines the acute effects of solid dark chocolate and liquid cocoa intake on endothelial function and blood pressure in overweight adults. DESIGN: Randomized, placebo-controlled, single-blind crossover trial of 45 healthy adults [mean age: 53 y; mean body mass index (in kg/m(2)): 30]. In phase 1, subjects were randomly assigned to consume a solid dark chocolate bar (containing 22 g cocoa powder) or a cocoa-free placebo bar (containing 0 g cocoa powder). In phase 2, subjects were randomly assigned to consume sugar-free cocoa (containing 22 g cocoa powder), sugared cocoa (containing 22 g cocoa powder), or a placebo (containing 0 g cocoa powder). RESULTS: Solid dark chocolate and liquid cocoa ingestion improved endothelial function (measured as flow-mediated dilatation) compared with placebo (dark chocolate: 4.3 +/- 3.4% compared with -1.8 +/- 3.3%; P < 0.001; sugar-free and sugared cocoa: 5.7 +/- 2.6% and 2.0 +/- 1.8% compared with -1.5 +/- 2.8%; P < 0.001). Blood pressure decreased after the ingestion of dark chocolate and sugar-free cocoa compared with placebo (dark chocolate: systolic, -3.2 +/- 5.8 mm Hg compared with 2.7 +/- 6.6 mm Hg; P < 0.001; and diastolic, -1.4 +/- 3.9 mm Hg compared with 2.7 +/- 6.4 mm Hg; P = 0.01; sugar-free cocoa: systolic, -2.1 +/- 7.0 mm Hg compared with 3.2 +/- 5.6 mm Hg; P < 0.001; and diastolic: -1.2 +/- 8.7 mm Hg compared with 2.8 +/- 5.6 mm Hg; P = 0.014). Endothelial function improved significantly more with sugar-free than with regular cocoa (5.7 +/- 2.6% compared with 2.0 +/- 1.8%; P < 0.001). CONCLUSIONS: The acute ingestion of both solid dark chocolate and liquid cocoa improved endothelial function and lowered blood pressure in overweight adults. Sugar content may attenuate these effects, and sugar-free preparations may augment them.